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Sécurités des Réseaux

PROJET DE FIN DE SESSION

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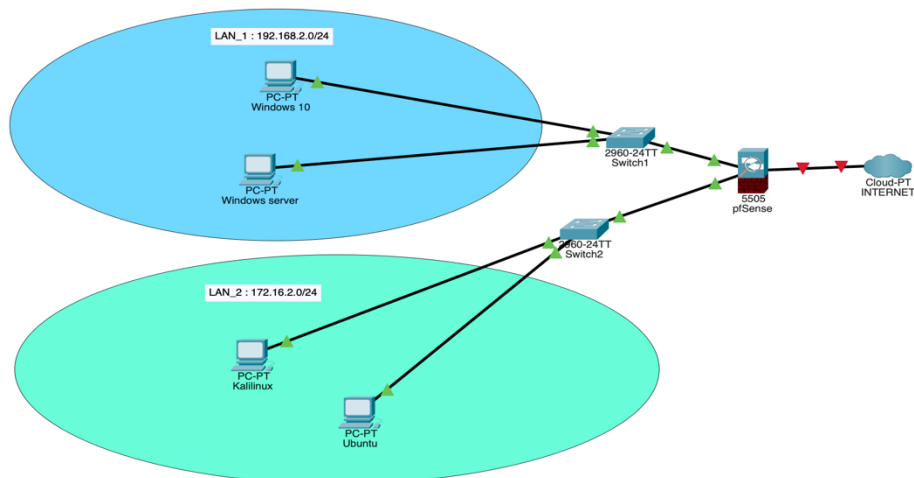
INTRODUCTION

Un pare-feu est un appareil de sécurité des réseaux qui contrôle le trafic réseau entrant et sortant et décide s'il bloque ou autorise le trafic selon un ensemble défini de règles de sécurité¹. Dans le cadre du projet de fin de session, il nous est demandé de créer dans VirtualBox un réseau de base comportant un réseau WAN et deux réseaux LANs et une machine virtuelle pfSense comme pare-feu avec trois cartes réseaux. En voici les résultats

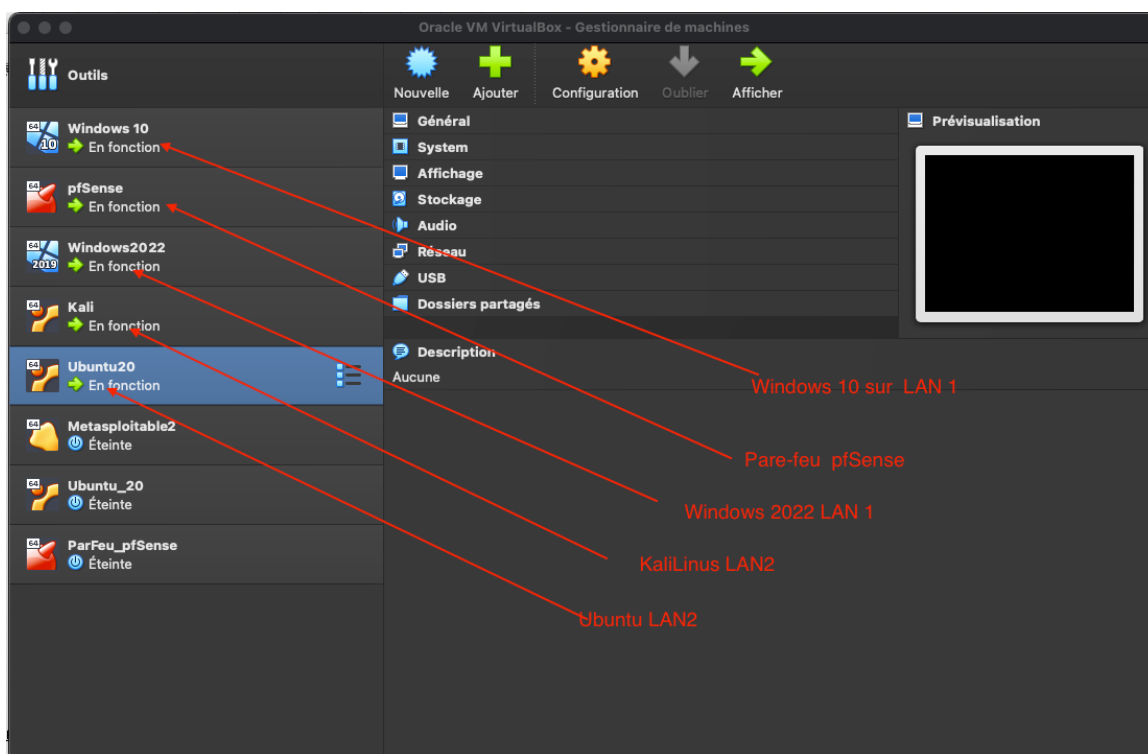
¹ Site Web CISCO : https://www.cisco.com/c/fr_ca/products/security/firewalls/what-is-a-firewall.html

TOPOLOGIE

Ci-dessous la topologie du réseau de notre travail



Représentation de nos différents VM dans VirtualBox



CONFIGURATION DE BASE PFSense

- En mode terminal

Choix de Lan à configurer

```
*** Welcome to pfSense 2.7.1-RELEASE (amd64) on pare-feu_pfsense ***

WAN (wan)      -> em0      -> v4/DHCP4: 192.168.1.102/24
LAN (lan)      -> em1      -> v4: 192.168.1.254/24
LAN2 (opt1)    -> em2      -> v4: 172.16.1.254/24

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults  13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                15) Restore recent configuration
7) Ping host                  16) Restart PHP-FPM
8) Shell

Enter an option: 2

Available interfaces:
1 - WAN (em0 - dhcp, dhcp6)
2 - LAN (em1 - static)
3 - LAN2 (em2 - static)

Enter the number of the interface you wish to configure: 2
```

Configuration de LAN_1 et de la passerelle

```
7) Ping host                  16) Restart PHP-FPM
8) Shell

Enter an option: 2

Available interfaces:
1 - WAN (em0 - dhcp, dhcp6)
2 - LAN (em1 - static)
3 - LAN2 (em2 - static)

Enter the number of the interface you wish to configure: 2

Configure IPv4 address LAN interface via DHCP? (y/n) n

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 192.168.2.254

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24
```

Configuration de DHCP sur le LAN2

```
Configure IPv4 address LAN interface via DHCP? (y/n) n

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 192.168.2.254

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via DHCP? (y/n) n

Enter the new LAN IPv6 address. Press <ENTER> for none:
>

Do you want to enable the DHCP server on LAN? (y/n) y
Enter the start address of the IPv4 client address range: 192.168.2.10
Enter the end address of the IPv4 client address range: 192.168.2.100
```

- En mode Web configuration

Activation de la troisième interface

Interfaces / Interface Assignments

Interface Assignments Interface Groups Wireless VLANs QinQs PPPs GREs GIGs Bridges LAGGs

Interface	Network port
WAN	em0 (08:00:27:9b:f2:4e)
LAN	em1 (08:00:27:7f:13:1d)
Available network ports:	em2 (08:00:27:49:30:8b)

Save Delete Add

Assignment de l'adresse IP

Interfaces / OPT1 (em2)

General Configuration

Enable ☒ Enable interface

Description LAN_2

IPv4 Configuration Type Static IPv4

IPv6 Configuration Type None

MAC Address xxxxxxxxxxxx

MTU 1500

MSS 60

Speed and Duplex Default (no preference, typically autoselect)

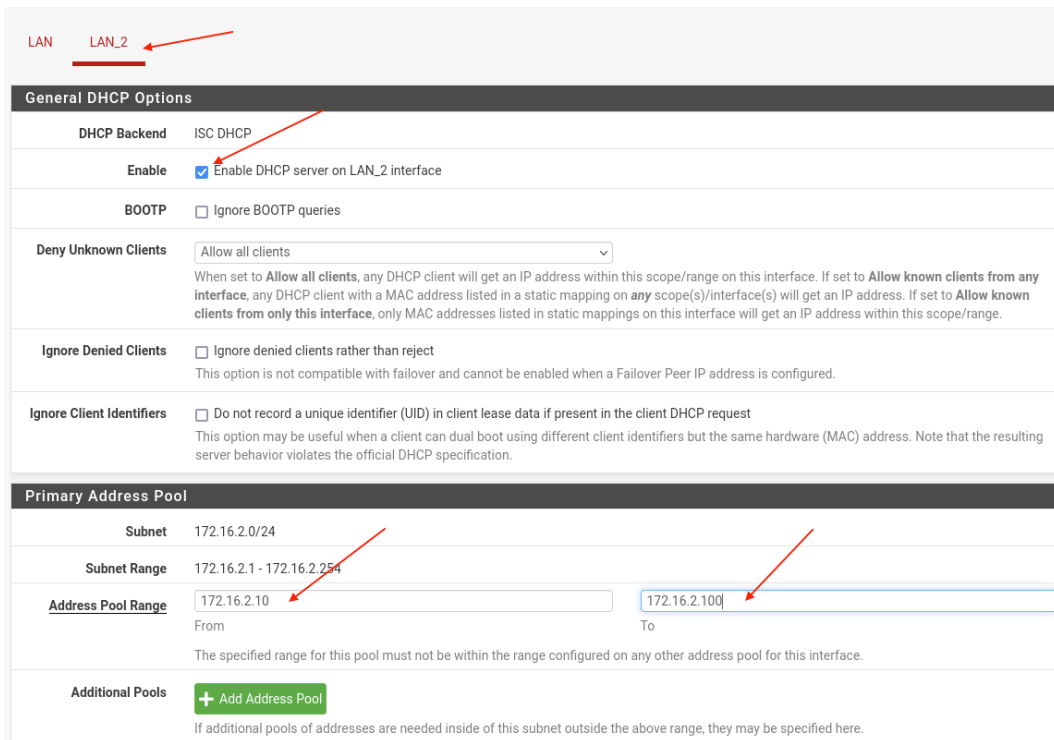
Static IPv4 Configuration

IPv4 Address 172.16.2.254 / 24

IPv4 Upstream gateway None Add a new gateway

Pour finir on clique sur sauvegarder

Configuration de DHCP sur le LAN_2 puis ajout de pool d'adresse début et fin



LAN **LAN_2**

General DHCP Options

DHCP Backend	ISC DHCP
Enable	☒ Enable DHCP server on LAN_2 interface
BOOTP	☐ Ignore BOOTP queries
Deny Unknown Clients	Allow all clients When set to Allow all clients, any DHCP client will get an IP address within this scope/range on this interface. If set to Allow known clients from any interface, any DHCP client with a MAC address listed in a static mapping on any scope(s)/interface(s) will get an IP address. If set to Allow known clients from only this interface, only MAC addresses listed in static mappings on this interface will get an IP address within this scope/range.
Ignore Denied Clients	☐ Ignore denied clients rather than reject This option is not compatible with failover and cannot be enabled when a Failover Peer IP address is configured.
Ignore Client Identifiers	☐ Do not record a unique identifier (UID) in client lease data if present in the client DHCP request This option may be useful when a client can dual boot using different client identifiers but the same hardware (MAC) address. Note that the resulting server behavior violates the official DHCP specification.

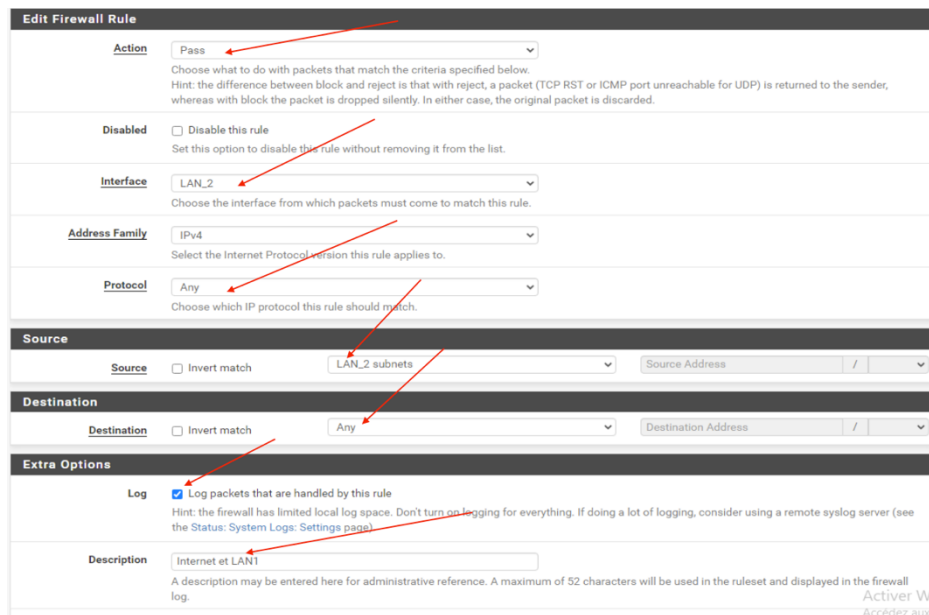
Primary Address Pool

Subnet	172.16.2.0/24
Subnet Range	172.16.2.1 - 172.16.2.254
Address Pool Range	From: 172.16.2.10 To: 172.16.2.100 The specified range for this pool must not be within the range configured on any other address pool for this interface.
Additional Pools	+ Add Address Pool If additional pools of addresses are needed inside of this subnet outside the above range, they may be specified here.

GESTION DES REGLES

Création de la règle

Cette règle nous permet d'avoir internet sur le LAN_2 et nous permet de communiquer avec le LAN_1

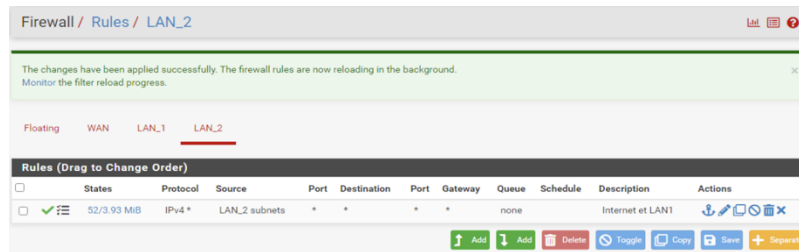


Edit Firewall Rule

Action	Pass	Choose what to do with packets that match the criteria specified below. Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.
Disabled	☐ Disable this rule	Set this option to disable this rule without removing it from the list.
Interface	LAN_2	Choose the interface from which packets must come to match this rule.
Address Family	IPv4	Select the Internet Protocol version this rule applies to.
Protocol	Any	Choose which IP protocol this rule should match.
Source	☐ Invert match	LAN_2 subnets
Destination	☐ Invert match	Any
Extra Options	☒ Log	Log packets that are handled by this rule. Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the Status: System Logs: Settings page).
Description	Internet et LAN1 A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset and displayed in the firewall log.	

Activer W
Accédez aux

La règle active



Test de connexion à partir du LAN2

Ubuntu étant sur le lan2, nous allons pinguer le serveur de google pour vérifier si on peut aller sur internet.

```
claude@Ubuntu20: ~  
claude@Ubuntu20:~$ ping 8.8.8.8  
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:  
64 bytes from 8.8.8.8: icmp_seq=1 ttl=106 time=52.6 ms  
64 bytes from 8.8.8.8: icmp_seq=2 ttl=106 time=29.5 ms  
64 bytes from 8.8.8.8: icmp_seq=3 ttl=106 time=81.4 ms  
64 bytes from 8.8.8.8: icmp_seq=4 ttl=106 time=82.3 ms  
64 bytes from 8.8.8.8: icmp_seq=5 ttl=106 time=26.3 ms  
64 bytes from 8.8.8.8: icmp_seq=6 ttl=106 time=26.2 ms  
64 bytes from 8.8.8.8: icmp_seq=7 ttl=106 time=26.8 ms  
64 bytes from 8.8.8.8: icmp_seq=8 ttl=106 time=34.5 ms  
64 bytes from 8.8.8.8: icmp_seq=9 ttl=106 time=43.5 ms  
64 bytes from 8.8.8.8: icmp_seq=10 ttl=106 time=30.8 ms  
64 bytes from 8.8.8.8: icmp_seq=11 ttl=106 time=69.7 ms  
64 bytes from 8.8.8.8: icmp_seq=12 ttl=106 time=45.4 ms  
64 bytes from 8.8.8.8: icmp_seq=13 ttl=106 time=36.5 ms  
64 bytes from 8.8.8.8: icmp_seq=14 ttl=106 time=24.7 ms  
64 bytes from 8.8.8.8: icmp_seq=15 ttl=106 time=34.3 ms  
64 bytes from 8.8.8.8: icmp_seq=16 ttl=106 time=54.7 ms  
64 bytes from 8.8.8.8: icmp_seq=17 ttl=106 time=31.1 ms  
64 bytes from 8.8.8.8: icmp_seq=18 ttl=106 time=51.6 ms  
64 bytes from 8.8.8.8: icmp_seq=19 ttl=106 time=33.2 ms
```

A partir de Kali Linux qui est sur le lan2 nous allons pinguer un poste de travail sur le lan1 (Windows 10)

```
claude@kali: ~  
File Actions Edit View Help  
(claude@kali)-[~]  
$ ping 192.168.2.11  
PING 192.168.2.11 (192.168.2.11) 56(84) bytes of data:  
64 bytes from 192.168.2.11: icmp_seq=1 ttl=127 time=2.42 ms  
64 bytes from 192.168.2.11: icmp_seq=2 ttl=127 time=1.51 ms  
64 bytes from 192.168.2.11: icmp_seq=3 ttl=127 time=1.36 ms  
64 bytes from 192.168.2.11: icmp_seq=4 ttl=127 time=1.38 ms  
64 bytes from 192.168.2.11: icmp_seq=5 ttl=127 time=1.50 ms  
64 bytes from 192.168.2.11: icmp_seq=6 ttl=127 time=1.77 ms  
64 bytes from 192.168.2.11: icmp_seq=7 ttl=127 time=1.54 ms  
64 bytes from 192.168.2.11: icmp_seq=8 ttl=127 time=2.28 ms  
64 bytes from 192.168.2.11: icmp_seq=9 ttl=127 time=1.58 ms  
64 bytes from 192.168.2.11: icmp_seq=10 ttl=127 time=1.47 ms  
64 bytes from 192.168.2.11: icmp_seq=11 ttl=127 time=1.62 ms  
64 bytes from 192.168.2.11: icmp_seq=12 ttl=127 time=1.88 ms  
64 bytes from 192.168.2.11: icmp_seq=13 ttl=127 time=2.84 ms
```

CONFIGURATION DU PORTAIL CAPTIF

Saisie du nom du portail captif

pfSense COMMUNITY EDITION System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

WARNING: The 'admin' account password is set to the default value. [Change the password in the User Manager.](#)

Services / Captive Portal / Add Zone

Add Captive Portal Zone

Zone name	Portail_Lan1
Zone name. Can only contain letters, digits, and underscores (_) and may not start with a digit.	
Zone description	Portail captif sur le lan 1
A description may be entered here for administrative reference (not parsed).	

Activation du portail captif et choix du LAN_1

pfSense COMMUNITY EDITION System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

WARNING: The 'admin' account password is set to the default value. [Change the password in the User Manager.](#)

Services / Captive Portal / Portail_Lan1 / Configuration

Configuration MACs Allowed IP Addresses Allowed Hostnames Vouchers High Availability File Manager

Captive Portal Configuration

Enable	☒ Enable Captive Portal
Description	Portail captif sur le lan 1
A description may be entered here for administrative reference (not parsed).	
Interfaces	☒ WAN ☒ LAN_1 ☐ LAN_2
Select the interface(s) to enable for captive portal.	

Choix du serveur d'authentification : Base de données locales

Authentication

Authentication Method	Use an Authentication backend ▾ Select an Authentication Method to use for this zone. One method must be selected. - "Authentication backend" will force the login page to be displayed and will authenticate users using their login and password, or using vouchers. - "None" method will force the login page to be displayed but will accept any visitor that clicks the "submit" button. - "RADIUS MAC Authentication" method will try to authenticate devices automatically with their MAC address without displaying any login page.
Authentication Server	☒ Local Database You can add a remote authentication server in the User Manager. Vouchers could also be used, please go to the Vouchers Page to enable them.
Secondary authentication Server	☐ Local Database You can optionally select a second set of servers to to authenticate users. Users will then be able to login using separated HTML inputs. This setting is useful if you want to provide multiple authentication method to your users. If you don't need multiple authentication method, then leave this setting empty.

Activation de la fenêtre de déconnection du portail captif

Logout popup window

☒ Enable logout popup window

If enabled, a popup window will appear when clients are allowed through the captive portal. This allows clients to explicitly disconnect themselves before the idle or hard timeout occurs.

Création des utilisateurs

WARNING: The 'admin' account password is set to the default value. [Change the password in the User Manager.](#)

System / User Manager / Users / Edit

UsersGroupsSettingsAuthentication Servers

User Properties

Defined by

USER

Disabled

☐ This user cannot login

Username

steve

Password

••••••

••••••

Full name

Steve-Anthony

User's full name, for administrative information only

Expiration date

12/31/2024

Leave blank if the account shouldn't expire, otherwise enter the expiration date as MM/DD/YYYY

Custom Settings

☐ Use individual customized GUI options and dashboard layout for this user.

Group membership

admins

Not member of

Member of

Move to "Member of" list

Move to "Not member of" list

Hold down CTRL (PC)/COMMAND (Mac) key to select multiple items.

Ajout de privilège aux utilisateurs

WARNING: The 'admin' account password is set to the default value. [Change the password in the User Manager.](#)

System / User Manager / Users / Edit / Add Privileges

UsersGroupsSettingsAuthentication Servers

User Privileges

User

steve (Steve-Anthony)

Assigned privileges

System - HA node sync

User - Config: Deny Config Write

User - Notices: View

User - Notices: View and Clear

User - Notices: Send login popup

User - System: Copy files (scp)

User - System: Copy files to home directory (chronos scp)

User - System: Shell account access

User - System: SSH forwarding

User - VPN: (Psec xauth Dialin

User - VPN: L2TP Dialin

User - VPN: PPTP Dialin

WebCity - AJAX: Get Queue Stats

WebCity - AJAX: Get Service Providers

WebCity - AJAX: Get Stats

WebCity - All pages

WebCity - Crash reporter

WebCity - Dashboard (all)

WebCity - Dashboard widgets (direct access)

WebCity - Diagnostics: AMP Table

Filter

Show only the choices containing this term

Privilege information

The following privileges effectively give the user administrator-level access because the user gains access to execute general commands, edit system

12 août 2024

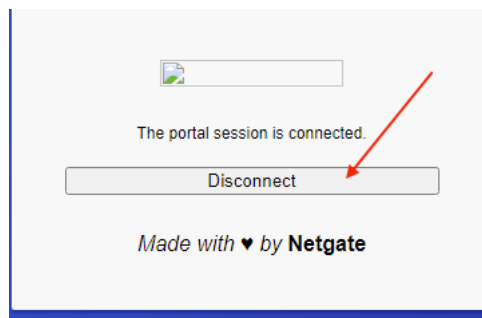
10

Test d'utilisation du portail captif

- Essai de connexion sur internet puis ouverture d'une fenêtre d'authentification d'utilisateur sur le portail captif

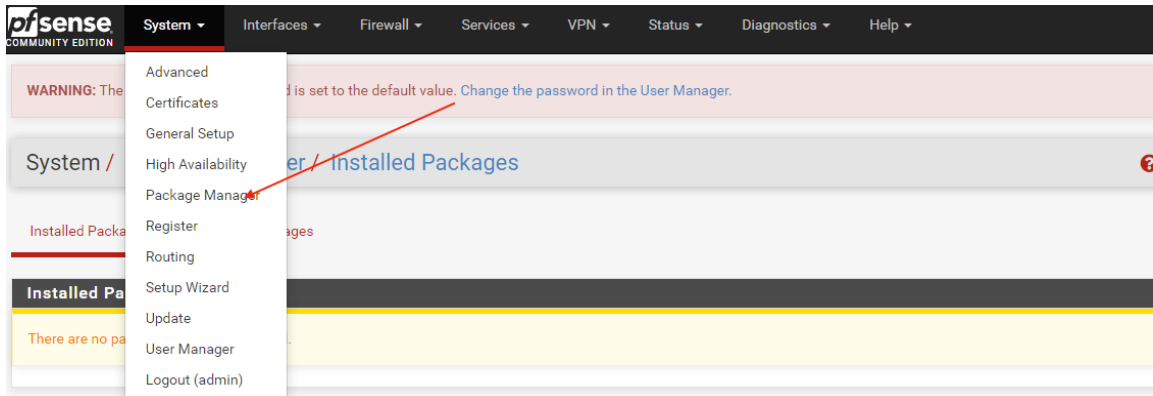


- Après la connexion nous avons une fenêtre de déconnexion

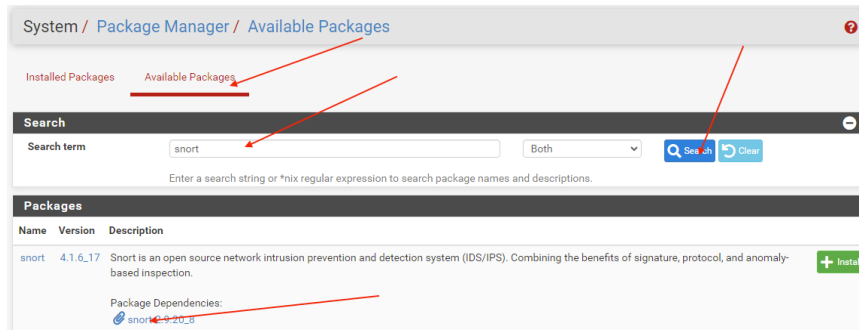


MISE EN PLACE DE IPS – SNORT

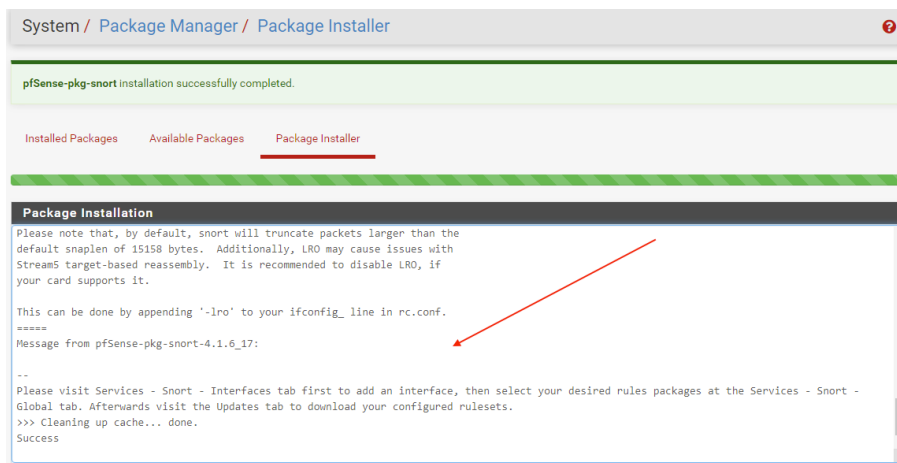
- Choisir le gestionnaire des paquets



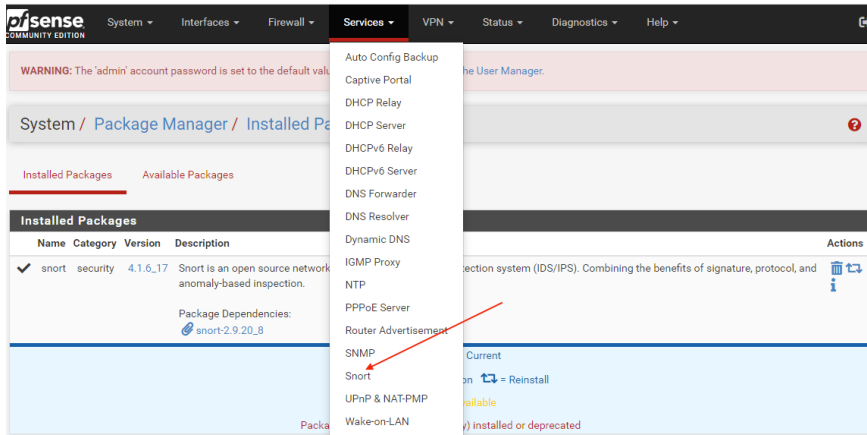
Choix des paquets disponibles pour installation



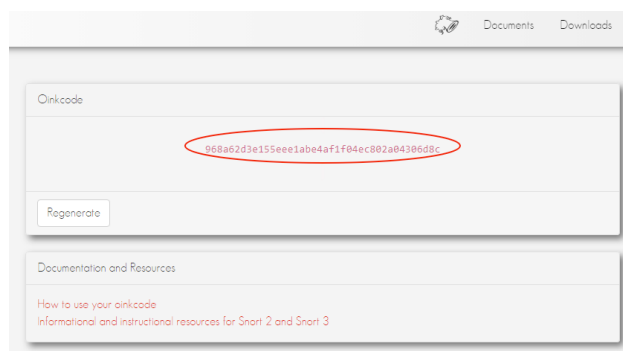
Installation de Snort



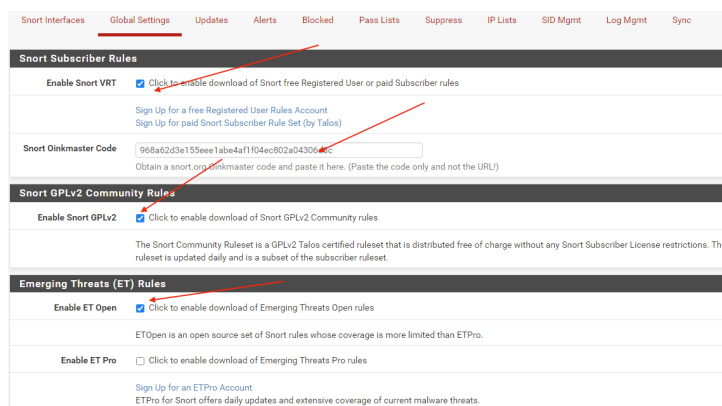
Choix de SNORT pour la configuration



Ajout de Oinkcode



Configuration de SNORT



Configuration de temps de Mise à jour

Rules Update Settings

Update Interval

1 DAY

Please select the interval for rule updates. Choosing NEVER disables auto-updates.

Update Start Time

00:05

Enter the rule update start time in 24-hour format (HH:MM). Default is 00 hours with a randomly chosen minutes value. Rules will update at the interval chosen above starting at the time specified here. For example, using a start time of 00:08 and choosing 12 Hours for the interval, the rules will update at 00:08 and 12:08 each day. The randomized minutes value should be retained to minimize the impact to the rules update site from large numbers of simultaneous requests.

Hide Deprecated Rules Categories

☒

Click to hide deprecated rules categories in the GUI and remove them from the configuration. Default is not checked.

Disable SSL Peer Verification

☐

Click to disable verification of SSL peers during rules updates. This is commonly needed only for self-signed certificates. Default is not checked.

General Settings

Remove Blocked Hosts Interval

NEVER

Please select the amount of time you would like hosts to be blocked. In most cases, one hour is a good choice.

Remove Blocked Hosts After Deinstall

☐

Click to clear all blocked hosts added by Snort when removing the package. Default is checked.

Keep Snort Settings After Deinstall

☒

Click to retain Snort settings after package removal.

Mise à jour en cours

WARNING: The 'admin' account password is set to the default value.

Services / Snort / Updates

Snort InterfacesGlobal SettingsUpdatesAlerts

Installed Rule Set MD5 Signature

Rule Set Name/Publisher	MD5 Signature Hash	MD5 Signature Date
Snort Subscriber Ruleset	798eb852be458bd270feef026956a5fb	Not Downloaded
Snort GPLv2 Community Rules	c02ca23ab2615cee21743775bc5a1ddf	Not Downloaded
Emerging Threats Open Rules	149fe9ac16f8f72e3429b77a606647b	Not Downloaded
Snort OpenAppID Detectors	Not Enabled	Not Enabled
Snort AppID Open Text Rules	Not Enabled	Not Enabled
Feodo Tracker Botnet C2 IP Rules	51b5326de130e1e0db29e5458fe944c2	Not Downloaded

Rules Update Task

Updating rule sets may take a while ... please wait for the process to complete.
This dialog will auto-close when the update is finished.

Close

IP ListsSID MgmtLog MgmtSync

MD5 Signature Date

Not Downloaded
Not Downloaded
Not Downloaded
Not Enabled
Not Enabled
Not Downloaded

Fin de Mise à jour

Installed Rule Set MD5 Signature		
Rule Set Name/Publisher	MD5 Signature Hash	MD5 Signature Date
Snort Subscriber Ruleset	798eb852be458bd270feef026956a5fb	Saturday, 10-Aug-24 15:54:14 EDT
Snort GPLv2 Community Rules	c02ca23ab2615cee21743775bc5a1ddf	Saturday, 10-Aug-24 15:54:14 EDT
Emerging Threats Open Rules	149fe9ac16f8f72e3429b77a606647b	Saturday, 10-Aug-24 15:54:14 EDT
Snort OpenAppID Detectors	Not Enabled	Not Enabled
Snort AppID Open Text Rules	Not Enabled	Not Enabled
Feodo Tracker Botnet C2 IP Rules	51b5326de130e1e0db29e5458fe944c2	Saturday, 10-Aug-24 15:54:04 EDT

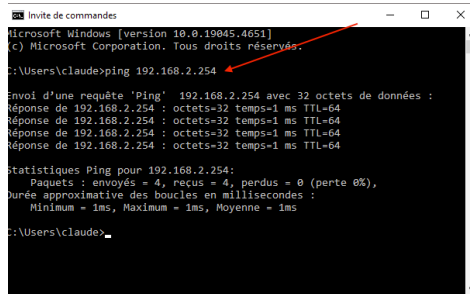
Update Your Rule Set		
Last Update	Aug-10 2024 15:54	Result: Success
Update Rules	Update Rules	Force Update
Click UPDATE RULES to check for and automatically apply any new posted updates for selected rules packages. Clicking FORCE UPDATE will zero out the MD5 hashes and force the download and application of the latest versions of the enabled rules packages.		

12 août 2024

14

CtizePREUVES DE PING

- Le ping à partir de Windows 10 (LAN_1)



```
Invite de commandes
Microsoft Windows [version 10.0.19045.4651]
(c) Microsoft Corporation. Tous droits réservés.

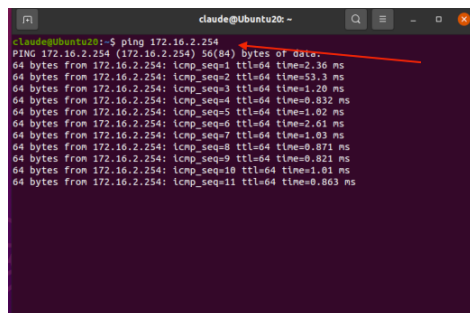
C:\Users\claud>ping 192.168.2.254

Envoi d'une requête 'Ping' 192.168.2.254 avec 32 octets de données :
Réponse de 192.168.2.254 : octets=32 temps=1 ms TTL=64
Réponse de 192.168.2.254 : octets=32 temps=1 ms TTL=64
Réponse de 192.168.2.254 : octets=32 temps=1 ms TTL=64
Réponse de 192.168.2.254 : octets=32 temps=1 ms TTL=64

Statistiques Ping pour 192.168.2.254:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
    Durée approximative des boucles en millisecondes :
        Minimum = 1ms, Maximum = 1ms, Moyenne = 1ms

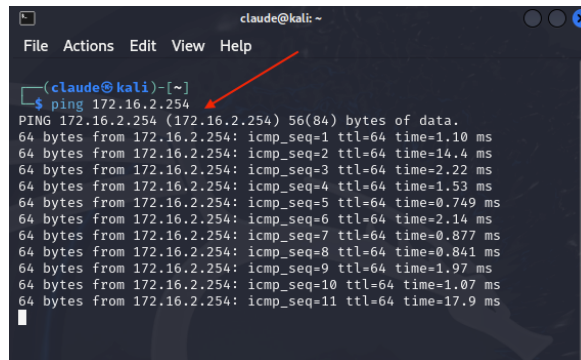
C:\Users\claud>
```

- Ping à partir de ubuntu (LAN_2)



```
claud@Ubuntu20:~$ ping 172.16.2.254
PING 172.16.2.254 (172.16.2.254) 56(84) bytes of data:
64 bytes from 172.16.2.254: icmp_seq=1 ttl=64 time=2.36 ms
64 bytes from 172.16.2.254: icmp_seq=2 ttl=64 time=59.3 ms
64 bytes from 172.16.2.254: icmp_seq=3 ttl=64 time=1.20 ms
64 bytes from 172.16.2.254: icmp_seq=4 ttl=64 time=0.832 ms
64 bytes from 172.16.2.254: icmp_seq=5 ttl=64 time=1.02 ms
64 bytes from 172.16.2.254: icmp_seq=6 ttl=64 time=2.61 ms
64 bytes from 172.16.2.254: icmp_seq=7 ttl=64 time=1.03 ms
64 bytes from 172.16.2.254: icmp_seq=8 ttl=64 time=0.871 ms
64 bytes from 172.16.2.254: icmp_seq=9 ttl=64 time=0.821 ms
64 bytes from 172.16.2.254: icmp_seq=10 ttl=64 time=1.81 ms
64 bytes from 172.16.2.254: icmp_seq=11 ttl=64 time=0.863 ms
```

- Ping à partir de KaliLinux (LAN_2)



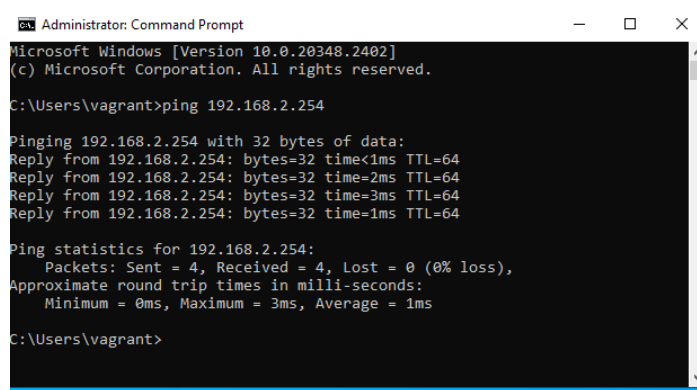
```

claude@kali: ~
File Actions Edit View Help

(claude@kali)-[~]
└─$ ping 172.16.2.254
PING 172.16.2.254 (172.16.2.254) 56(84) bytes of data:
64 bytes from 172.16.2.254: icmp_seq=1 ttl=64 time=1.10 ms
64 bytes from 172.16.2.254: icmp_seq=2 ttl=64 time=14.4 ms
64 bytes from 172.16.2.254: icmp_seq=3 ttl=64 time=2.22 ms
64 bytes from 172.16.2.254: icmp_seq=4 ttl=64 time=1.53 ms
64 bytes from 172.16.2.254: icmp_seq=5 ttl=64 time=0.749 ms
64 bytes from 172.16.2.254: icmp_seq=6 ttl=64 time=2.14 ms
64 bytes from 172.16.2.254: icmp_seq=7 ttl=64 time=0.877 ms
64 bytes from 172.16.2.254: icmp_seq=8 ttl=64 time=0.841 ms
64 bytes from 172.16.2.254: icmp_seq=9 ttl=64 time=1.97 ms
64 bytes from 172.16.2.254: icmp_seq=10 ttl=64 time=1.07 ms
64 bytes from 172.16.2.254: icmp_seq=11 ttl=64 time=17.9 ms

```

- Ping à partir de Windows server 2022 (LAN_1)



```

Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.2402]
(c) Microsoft Corporation. All rights reserved.

C:\Users\vagrant>ping 192.168.2.254

Pinging 192.168.2.254 with 32 bytes of data:
Reply from 192.168.2.254: bytes=32 time<1ms TTL=64
Reply from 192.168.2.254: bytes=32 time=2ms TTL=64
Reply from 192.168.2.254: bytes=32 time=3ms TTL=64
Reply from 192.168.2.254: bytes=32 time=1ms TTL=64

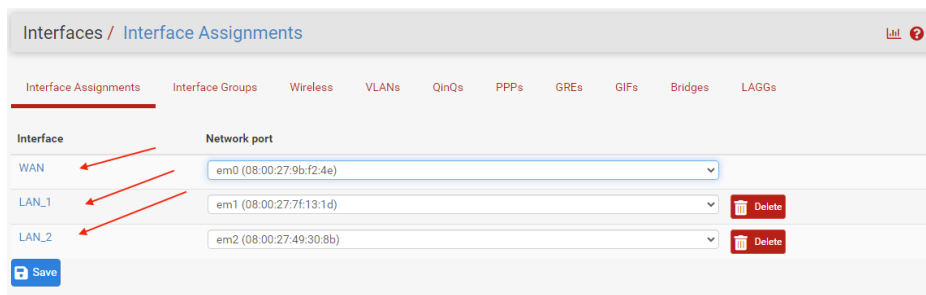
Ping statistics for 192.168.2.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\Users\vagrant>

```

CAPTURE DES TROIS INTERFACES DANS LE WEBCONFIGURATOR

Ci-dessous les trois interfaces : WAN, LAN_1 et LAN_2



PREUVE DE CREATION DE TROIS UTILISATEURS SUR LE LAN 1

Les trois utilisateurs que nous avons créés

WARNING: The 'admin' account password is set to the default value. [Change the password in the User Manager.](#)

System / User Manager / Users

Users

Groups

Settings

Authentication Servers

Users

	Username	Full name	Status	Groups	Actions
☐	Neomie	Néomie	✓		
☐	Tidiane	Tidiane Marega	✓		
☐	admin	System Administrator	✓	admins	
☐	steve	Steve-Anthony	✓		

+ Add

Delete

CONCLUSION

En somme, nous avons configuré les plages de DHCP sur pfSense en mode terminal et en mode configuration web. Nous avons par la suite créé des règles qui nous permettent d'aller sur internet à partir du LAN 2. Enfin nous avons configuré le portail captif et une configuration d'IPS SNORT.